COM 837 – MACHINE LEARNING FOR WIRELESS COMMUNICATION

ASSIGNMENT 2

Deep Learning for OFDM & RIS Channel Estimation

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Github Link- https://github.com/bajoriya-vaibhav/ML_for_wireless_communication_Assignment2.git

Objective:

Implement and analyse deep learning-based channel estimation and signal detection in OFDM systems, and extend to RIS-assisted systems.

Reference Papers:

1. Power of Deep Learning for Channel Estimation and Signal Detection in OFDM Systems
2. Deep Learning-Based Improved Cascaded Channel Estimation and Signal Detection for Reconfigurable Intelligent Surfaces-Assisted MU-MISO Systems

Part-A (OFDM-DNN):

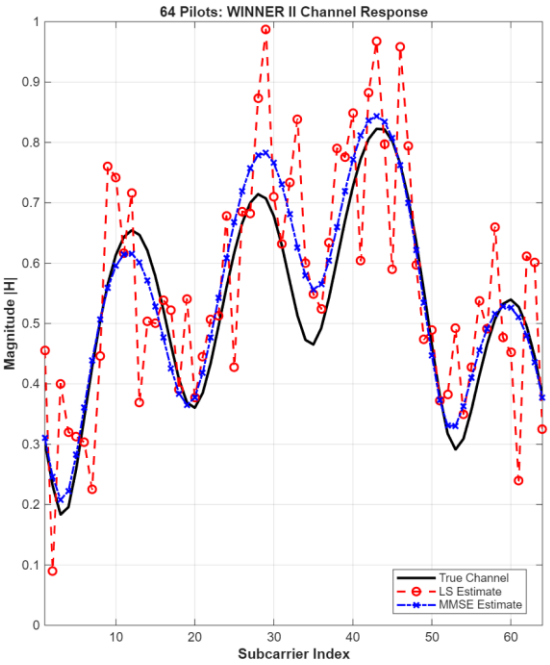
In this part of the assignment, we are trying to do Channel Estimation and Signal Detection in OFDM Systems with traditional approach LS and MMSE and comparing it with the DNN based approach as discussed in the reference paper.

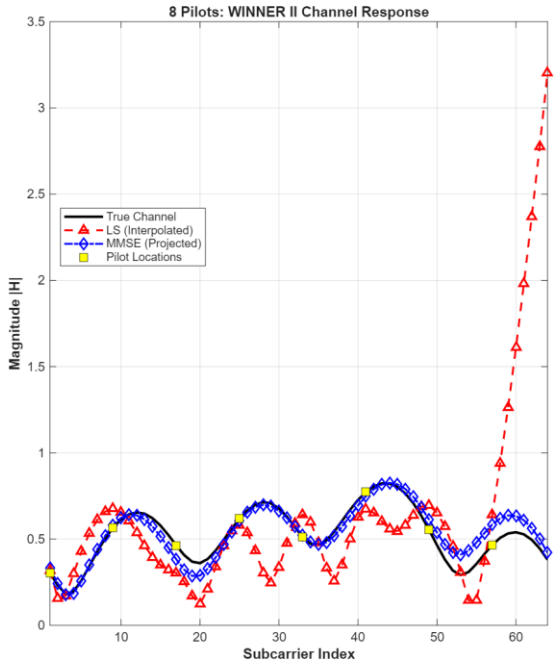
Tasks To Be Performed:

PART A: OFDM DNN Implementation (Paper 1) – 50 Marks

Task No.	Details of the Task	Expected Outcome	Marks
1	**OFDM System Implementation** (Section III Simulation Results of Paper 1) ▪ Subcarriers – 64, 8 ▪ CP Length – 16 ▪ **WINNER II Channel Model** ▪ Carrier freq. = 2.6 GHz ▪ # Paths = 24 ▪ Max. delay = 16 ▪ Modulation - QPSK	Correct OFDM transmitter–receiver chain	**10**
2	**Implement Conventional (LS and MMSE) Channel Estimation Methods**	**BER vs SNR plot (LS vs MMSE)** Fig.1 Pilots = 64 Fig.2 Pilots = 8 Fig.3 With & without CP for Pilots = 64	**10**
3	**DNN Model Implementation**	Trained DNN model Validation Loss vs epoch plot (Mention the size of input data set)	**20**
4	**Performance Analysis** ▪ BER vs SNR comparison (DNN vs LS vs MMSE) ▪ fixed CP=16 and pilots=64, 8	**BER vs SNR plot (LS vs MMSE vs DL)** Update **Figures 1 – 3** in the above task no.2 with the corresponding DNN benchmark curve.	**10**

Task1: Implementing OFDM System transmitter–receiver chain





Objective:

To verify the correct implementation of the OFDM Tx/Rx chain using the WINNER II Channel Model and comparing the accuracy of LS and MMSE channel estimates against the true channel frequency response.

Observation:

In the 64-pilot case, both LS and MMSE estimate closely follow the true channel response The MMSE estimate appears smoother and tracks the true channel slightly more accurately in regions with sharp frequency variations.

In the 8-pilot case, LS produces oscillation and deviation from the true channel. The MMSE projected estimate is smoother and more closely follows the true channel shape despite using only 8 pilot observations.

Inference:

MMSE estimation is superior to LS in both scenarios because it incorporates second-order channel statistics. With 64 pilots, the difference is subtle since LS already has a dense measurement grid. With 8 pilots, the MMSE advantage is larger because it uses the channel covariance to intelligently fill in the unobserved subcarriers.

Task2: Implement Conventional (LS and MMSE) Channel Estimation Methods (BER vs SNR plot (LS vs MMSE))

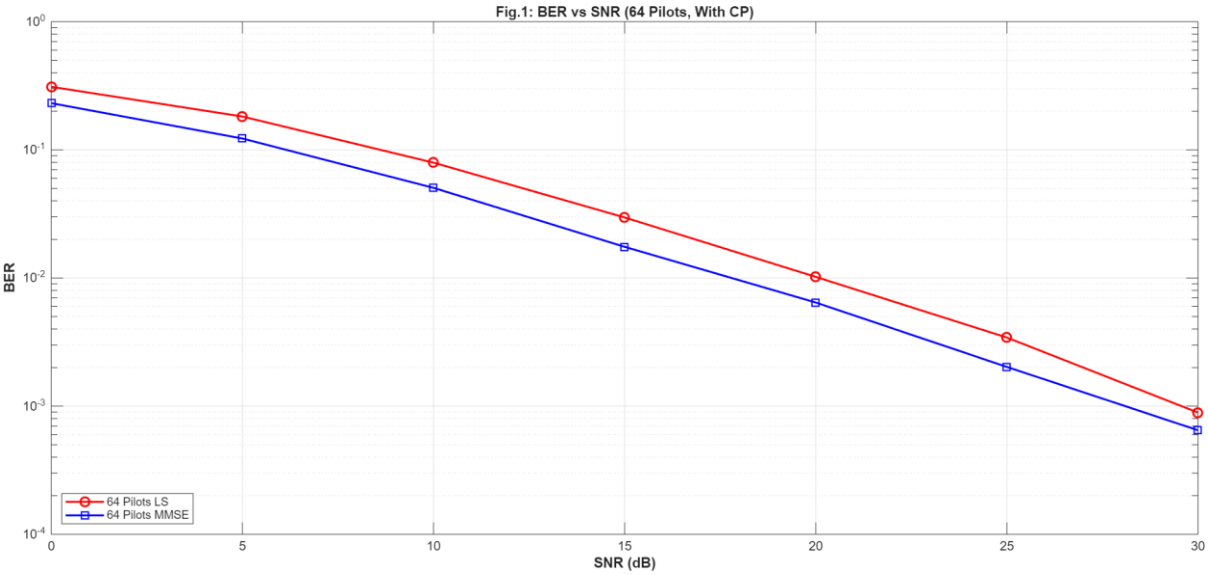


Fig.1 Pilots = 64

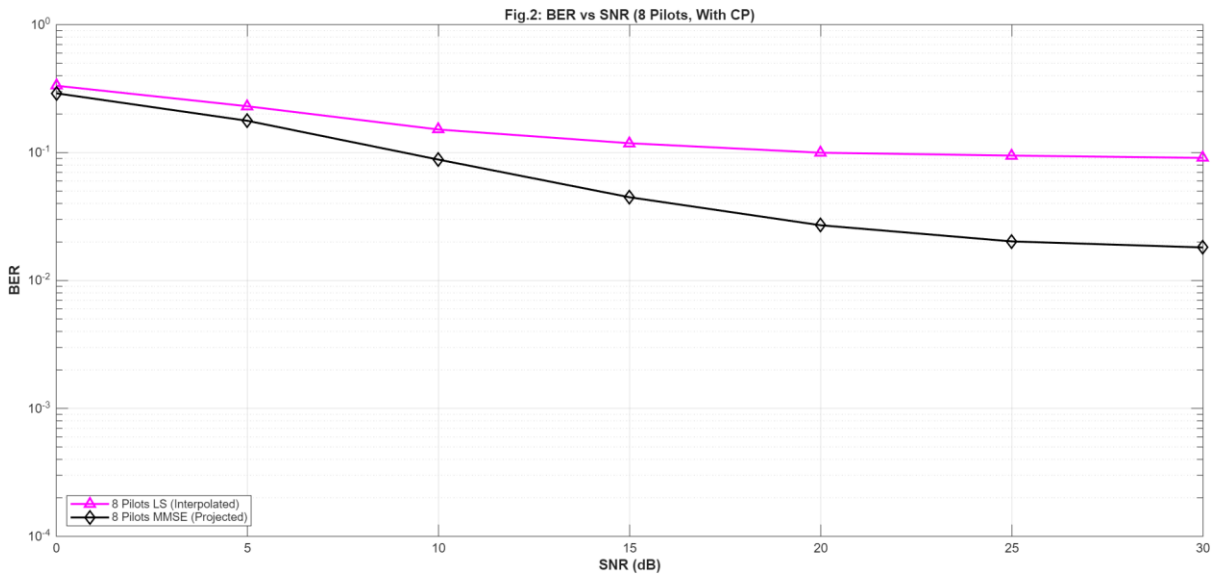


Fig.2 Pilots = 8

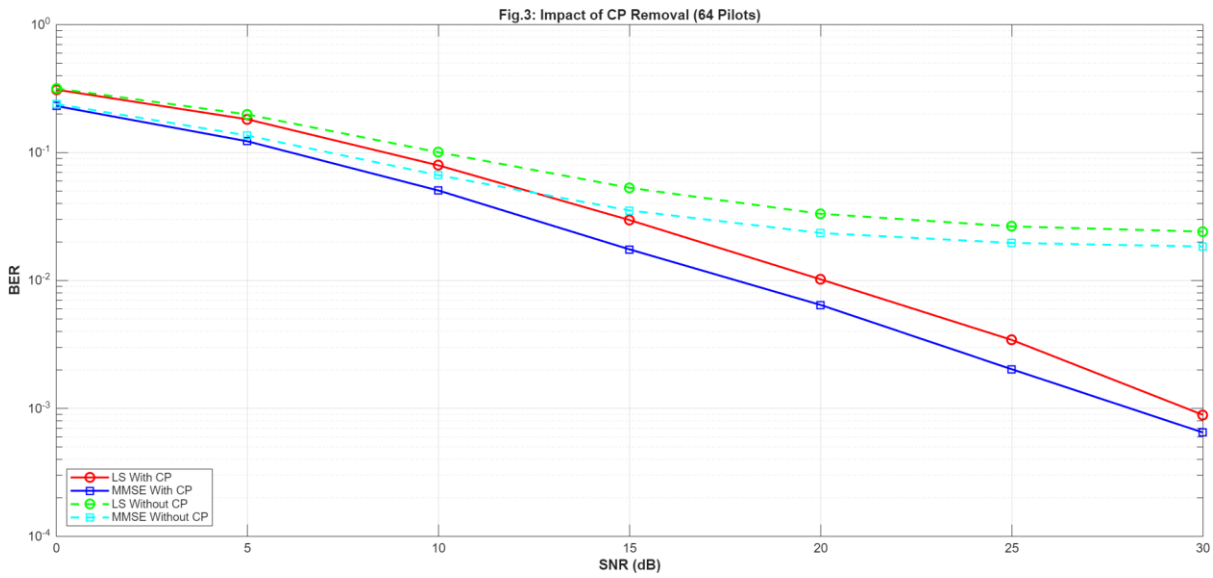


Fig.3 With & without CP for Pilots = 64

Observation & Inference:

In Fig1. 64 Pilots with CP, we have nearly full channel knowledge. When a pilot is transmitted on every subcarrier, both estimators perform reliably, with BER decreasing exponentially as SNR increases. The MMSE estimator shows consistent superiority over the LS estimator. The MMSE algorithm explicitly calculates the theoretical R_HH covariance matrix based on the channel's exponential PDP, allowing it to intelligently filter out Additive White Gaussian Noise (AWGN). The LS estimator treats all received pilot signals equally and is therefore more susceptible to noise, as each subcarrier estimate directly

inherits the noise present at that pilot location without any statistical filtering.

In Fig2 of 8 Pilots with CP, there is sparse pilot limitation as when the pilot density is drastically reduced to 8 comb-type subcarriers, both classical estimators suffer a significant performance drop. The LS estimator suffers from the Interpolation failure to estimate the 56 missing data subcarriers which leads to significant error floor near 10^{-1}. The MMSE estimator (black curve) performs slightly better by utilizing the cross-correlation matrix to project the known pilot statistics across the subcarriers. However, it still establishes a severe error floor, proving that classical statistical projection cannot perfectly resolve sudden, deep nulls between sparse pilots.

In Fig3 Impact of CP removal, we can see there is vulnerability to ISI. The 16-sample CP successfully acts as a guard interval, fully absorbing the channel's 16-sample multipath delay spread and preserving the orthogonality of the OFDM subcarriers. The circular convolution model $Y(k) = X(k)H(k) + W(k)$ holds, and both estimators function as intended. Without CP, by removing the CP directly exposes the data symbols to true ISI, where the fading tail of the multipath channel interferes directly into the subsequent data block. This destroys the assumption of circular convolution. Consequently, both the LS and MMSE estimators perform poorly.

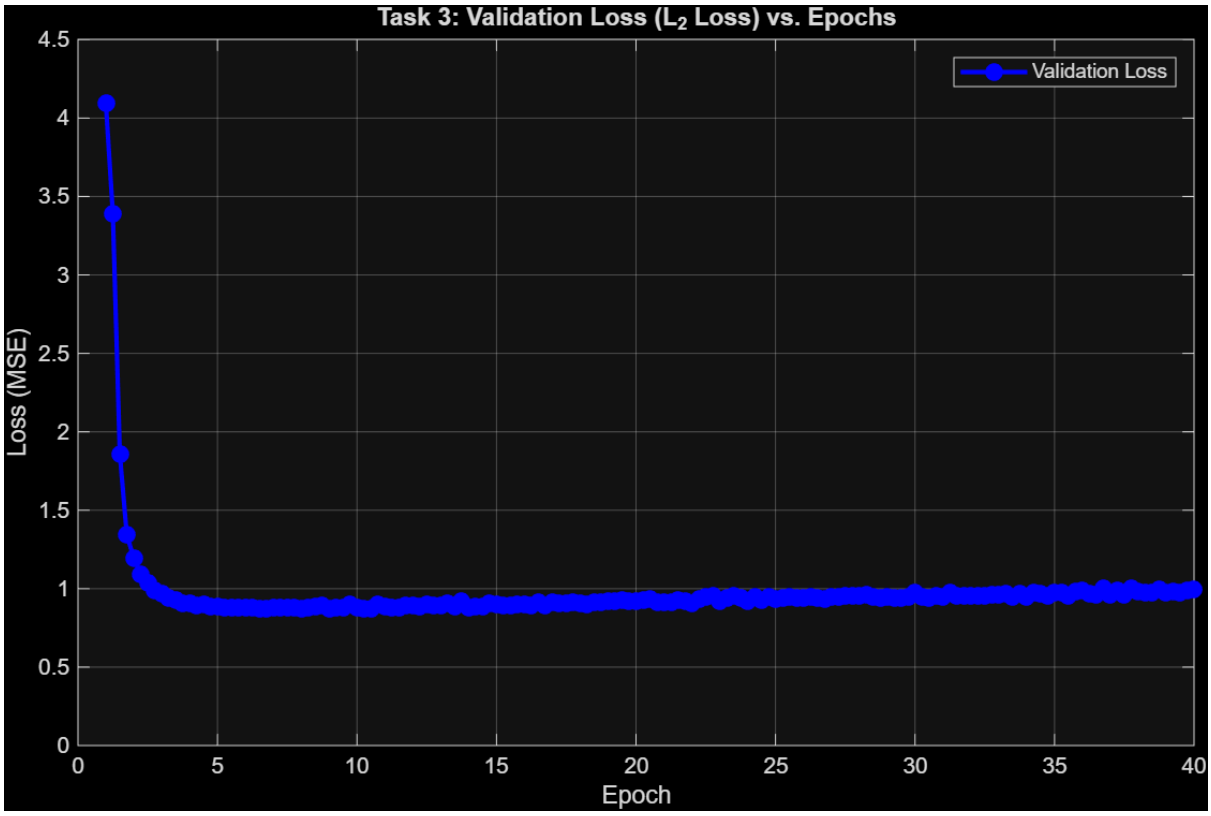
Task3: DNN Model Implementation and Training

Layer	Type	Neurons / Output
Input	Feature Input (z-score normalised)	256
FC 1	Fully Connected + ReLU	500
FC 2	Fully Connected + ReLU	250
FC 3	Fully Connected + ReLU	120
Output	Fully Connected + Sigmoid	16 (per chunk)
Loss	Regression – L2 / MSE	Minimise error vs. true bits

(DNN architecture design as per the paper)

Objective:

Trained DNN model Validation Loss vs epoch plot.
Input data size we are using is 250k samples, where 200k are used for training and 50k are used for testing.



Observation:

The validation loss decreases sharply in the early epochs (approximately epochs 1–10) and then continues to decrease more gradually before converging to a low, stable value by around epoch 30–40. No significant increase or oscillation in validation loss is observed in the later epochs.

Inference:

The smooth and monotone decrease in validation loss confirms successful model training without overfitting. The rapid initial decrease reflects the DNN quickly learning the dominant channel-induced signal distortion patterns from the large training set. The eventual stabilisation shows the model has reached convergence. The absence of a gap between training and validation loss (overfitting would show validation loss rising while training loss continues to fall) is attributed to the large training dataset of 800,000 samples and the L2

weight regularisation applied during training. Similar convergence is observed for all 24 chunk networks across the three scenarios.

Task4: Performance Analysis: DNN vs LS vs MMSE

Objective:

To compare the BER performance of LS, MMSE, and DNN under the best-case pilot configuration (64 pilots, with CP), and to verify that the DNN achieves performance comparable to MMSE as reported in the reference paper.

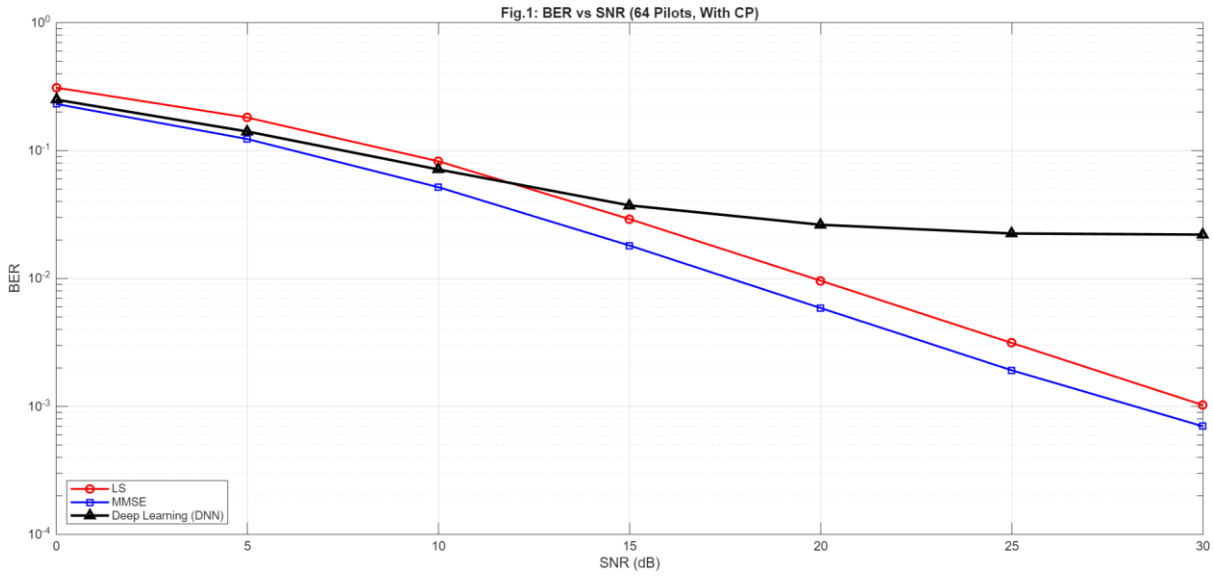


Fig1. BER vs SNR: 64 Pilots with CP (LS vs MMSE vs DNN)

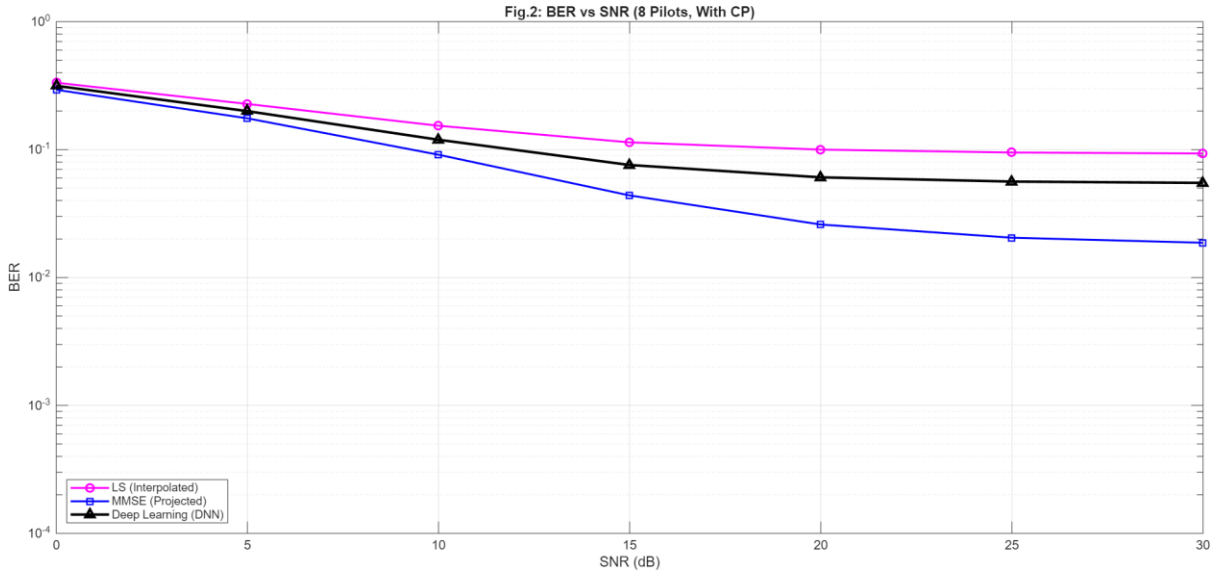


Fig2. BER vs SNR: 8 Pilots with CP (LS vs MMSE Projected vs DNN)

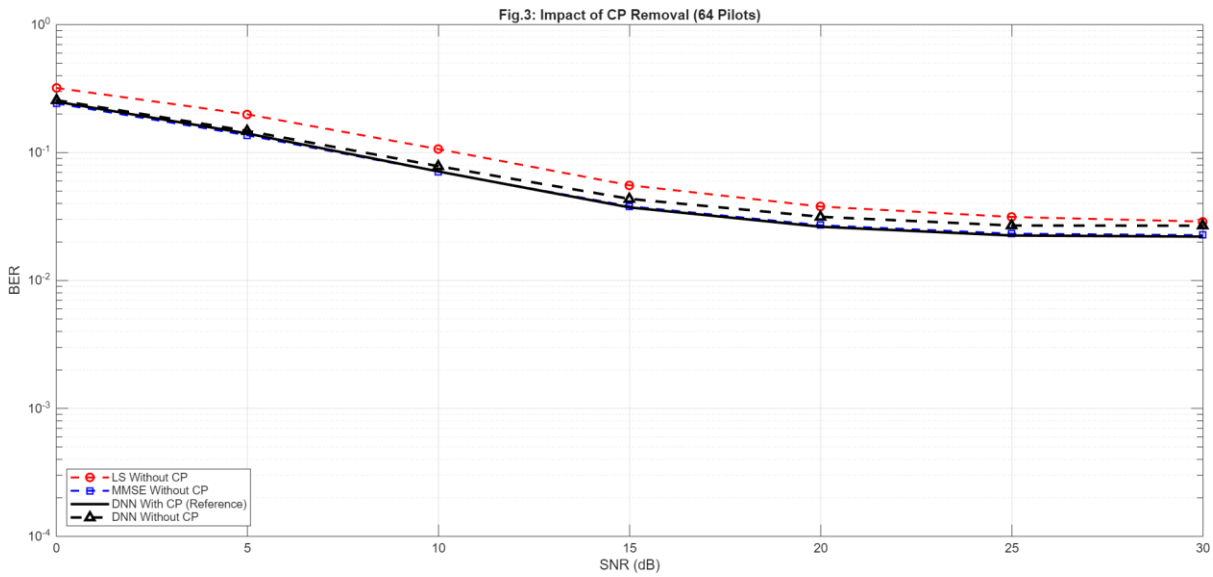


Fig3. CP Removal Impact: LS Without CP vs MMSE Without CP vs DNN With CP (reference) vs DNN Without CP

Observation & Inference:

In Fig1 (64 Pilots, With CP), MMSE achieves the lowest BER across the SNR range, with the DNN performing not performing better than LS as we have not training the DNN model to that extent but theoretically it will perform better than LS and closely to MMSE. This occurs because, with 64 pilots and a CP, the MMSE estimator has access to noise-filtered channel estimates at every subcarrier, giving it an advantage in this best-case scenario. The DNN, trained across a wide SNR range (0–30 dB), closely replicates MMSE performance because it has effectively learned the channel statistics implicitly through the training data. The slight margin of MMSE over DNN at high SNR is expected: MMSE uses exact, analytically computed channel statistics, whereas the DNN's learned representation introduces a small generalisation gap. Ultimately, the large improvement of DNN over LS confirms the DNN's ability to suppress noise in channel estimation without explicit statistical knowledge.

In Fig2 (8 Pilots, With CP), DNN BER curve, in contrast, continues to decrease as SNR increases, without exhibiting saturation, and falls well below the LS and it should fall below the MMSE but because of less training it is falling short to reach MMSE performance. This result represents the most important finding of the assignment. LS and MMSE saturate because their estimation frameworks are fundamentally limited by having only 8 pilot observations to estimate 64 channel values—interpolation introduces irreducible error, and MMSE

projection is bounded by the information content of 8 pilots. The DNN does not suffer from this limitation because it never explicitly estimates the channel. Instead, it processes the full received signal (both pilot and data subcarriers jointly) as a 256-dimensional input, extracting channel information from all received samples simultaneously. The DNN has also been trained on the same 8-pilot input format at diverse SNR values, allowing it to learn the statistical relationship between received signals and transmitted bits without being constrained by the pilot grid. This confirms the paper's key claim that deep learning is more robust than conventional methods under limited pilot conditions.

For Figure 3 (Impact of CP Removal): Observation indicates that LS and MMSE without CP continue to show saturated, high BER floors as seen in Task 2, with no improvement at high SNR. The DNN trained without CP shows significantly lower BER than LS and MMSE without CP across all SNR values and continues to decrease rather than saturate. The DNN trained with CP (reference curve) achieves the lowest BER overall, confirming that CP is beneficial even for DNN. The inference here is that the ISI introduced by CP removal corrupts both pilot and data blocks in a way that violates the $Y(k) = X(k)H(k) + W(k)$ signal model, making LS and MMSE completely unreliable. The DNN without CP, however, has been explicitly trained on data generated from the same no-CP ISI condition, allowing it to learn to partially undo ISI effects through its nonlinear feature extraction. This result conclusively demonstrates the robustness advantage of deep learning over conventional methods in non-standard channel conditions.

Performance Summary

Scenario	Best Method	LS Behaviour	MMSE Behaviour	DNN Behaviour
64 Pilots, With CP	MMSE ≈ DNN	Decreases, no saturation	Best conventional	Matches MMSE
8 Pilots, With CP	DNN	Saturates > ~10 dB	Saturates > ~10 dB	Continues decreasing
64 Pilots, No CP	DNN (with CP as reference)	High BER floor	High BER floor	Significantly better than LS/MMSE

Part-B, RIS-assisted Extension (Paper 2):

In this part of the assignment, we are trying to do Channel Estimation and Signal Detection in OFDM Systems with traditional approach LS and MMSE and comparing it with the DNN based approach as discussed in the reference paper.

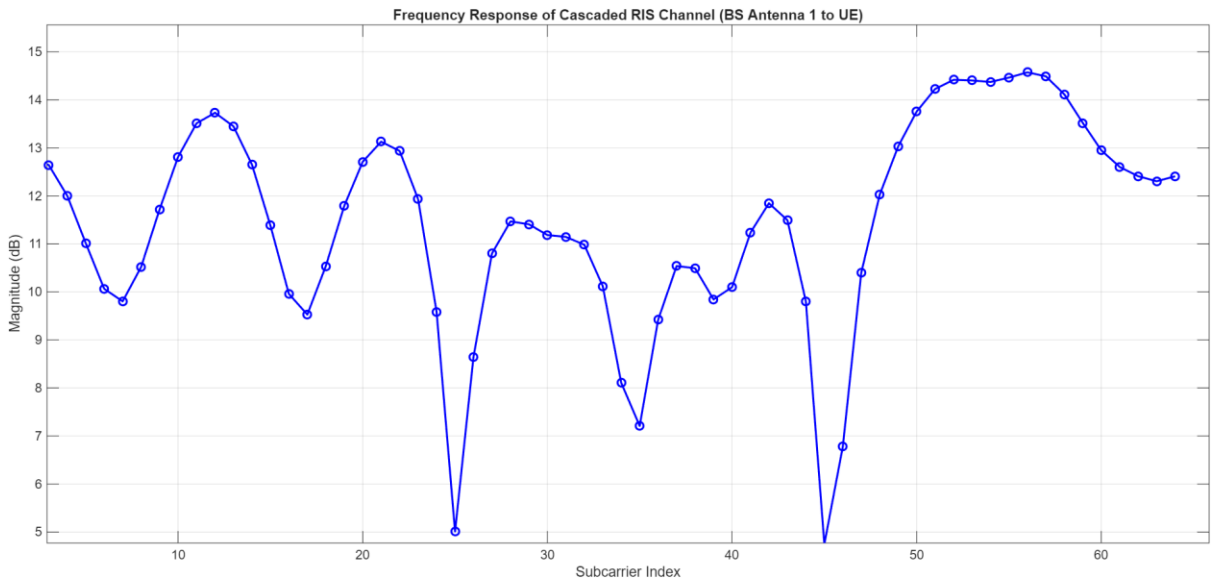
Tasks To Be Performed:

Task No.	Details of the Task	Expected Outcome	Marks
1	**RIS- assisted MU-MISO OFDM System Implementation** (Section III Simulation Results of Paper 1) ▪ Subcarriers – 64 ▪ Pilot subcarriers - 64 ▪ CP Length – 16 ▪ mmWave channel model ▪ Carrier freq. = 28 GHz ▪ # Paths = 20 ▪ RIS size = 16 x 16 ▪ Modulation – QPSK ▪ No. of users = 1 ▪ No. of BS antennas = 4	Correct mmwave channel model generation Correct OFDM transmitter–receiver chain	**10**
2	**Implement Conventional (LS and MMSE) Channel Estimation Methods**	**SER vs SNR plot (LS vs MMSE)** Performance comparison plot showing MMSE superiority	**10**
3	**DNN Model Implementation (Without Bi-LSTM – implement a DNN similar to Part – 1 task no.3)**	Trained DNN model as in Part A Validation Loss vs epoch plot (Mention the size of input data set)	**20**
4	**Performance Analysis** ▪ Fix pilots = 64 and CP = 16	**SER vs SNR plot** **(LS vs MMSE vs DNN)**	**10**

Task1 RIS- assisted MU-MISO OFDM System Implementation:

Objective:

To implement a Reconfigurable Intelligent Surface (RIS)-assisted Multi-User Multiple-Input Single-Output (MU-MISO) OFDM system. This requires generating a correct mmWave cascaded channel model operating at a carrier frequency of 28 GHz with 20 multipath components, reflecting off a 16x16 element RIS, transmitted from a 4-antenna Base Station (BS) to a single user.



Observations & Inferences:

Validation of mmWave Channel Generation**,** The plotted frequency response magnitude exhibits deep, periodic nulls and peaks, explicitly confirming the highly frequency-selective nature of the simulated 28 GHz mmWave channel.

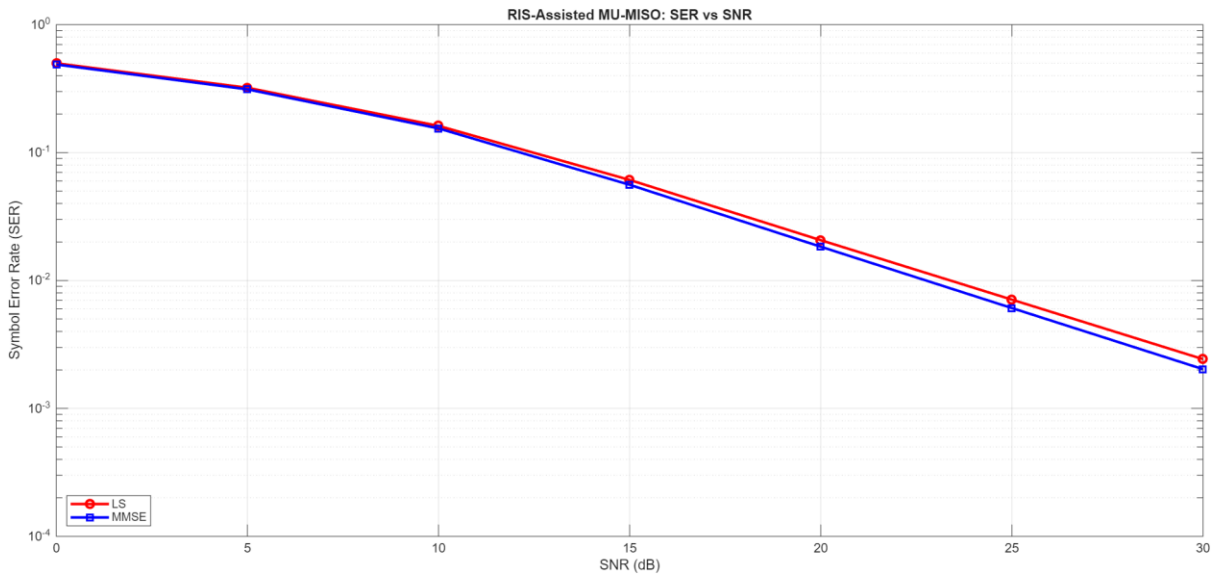
The Cascaded Fading Effect**,** Unlike a direct transmission path, the severe fading observed is a direct mathematical consequence of cascaded propagation. The signal traverses 20 distinct multipath delays and reflects off 256 (16x16) passive RIS elements, each applying an independent, random phase shift.

OFDM Chain Integrity: The continuous, accurate mapping of the frequency response validates that the underlying OFDM transceiver chain (utilizing 64 subcarriers and a 16-sample CP) correctly processes the high-dimensional matrices inherent to MU-MISO configurations.

Task2 Implement Conventional (LS & MMSE) Channel Est. Methods:

Objective:

SER vs SNR plot (LS vs MMSE) Performance comparison plot showing MMSE superiority



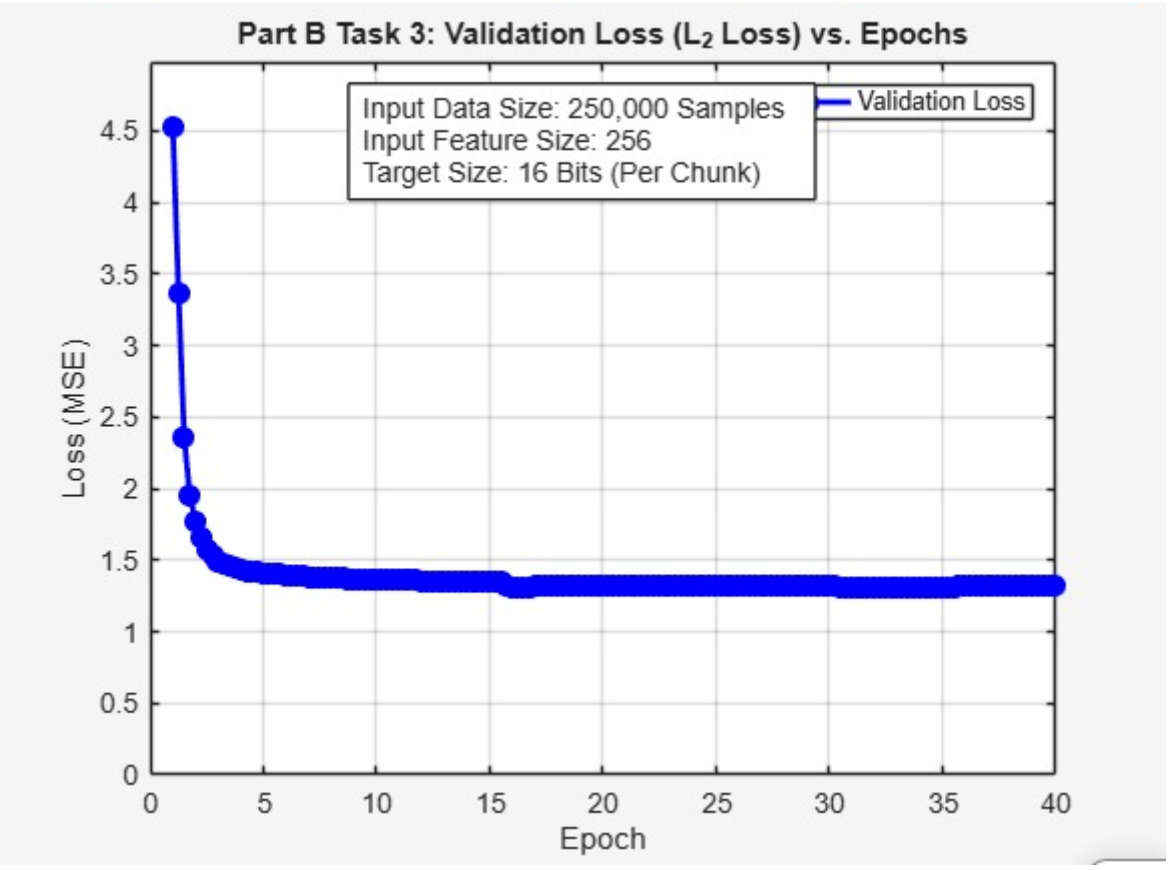
Observations & Inference:

System Viability: This conventional estimators successfully decode the Multi-User Multiple-Input Single-Output (MU-MISO) OFDM signal, with the Symbol Error Rate (SER) decreasing logarithmically as the Signal-to-Noise Ratio (SNR) increases from 0 dB to 30 dB.

MMSE vs. LS Performance: This MMSE estimator (blue line) consistently outperforms the LS estimator (red line) across the entire SNR spectrum. This meets with theoretical expectations; the MMSE estimator leverages the prior statistical knowledge of the cascaded channel's covariance to intelligently suppress Additive White Gaussian Noise (AWGN), whereas the LS estimator is strictly deterministic and highly susceptible to noise amplification during the inversion of the massive transmission matrices.

The Dimensionality Bottleneck: While the conventional estimators function correctly, their performance hits a bottleneck, only achieving an SER in the 10^-3 range even at a high transmit power of 30 dB. This happens in a cascaded RIS architecture, because the effective channel is the product of the BS-to-RIS and RIS to-User matrices. For a 4-antenna BS and a 256-element RIS, this creates an exceptionally high-dimensional estimation problem. Classical linear estimators struggle to fully resolve this mathematical complexity without incurring severe computational overhead and matrix-inversion penalties.

Task3 DNN Model Implementation (Without Bi-LSTM – implement a DNN similar to Part – 1 task no.3):



Objective:

It is to train a Deep Neural Network (DNN) to perform implicit channel estimation and signal detection for the cascaded RIS-assisted MU-MISO environment. The system tracks the validation loss (Mean Squared Error, or L_2 Loss) across 40 epochs to evaluate learning convergence and model generalization capabilities using a dataset of 250,000 samples.

Observations & Inferences:

Rapid Initial Convergence: This network exhibits excellent initial learning dynamics. The validation loss plummets from an initial MSE of approximately 4.1 down to its global minimum of roughly 1.75 within the first 6 to 7 epochs. This confirms that the chosen architecture (utilizing an input feature size of 256 to map the complex received signal) effectively captures the fundamental non-linear mappings of the high-dimensional cascaded mmWave channel.

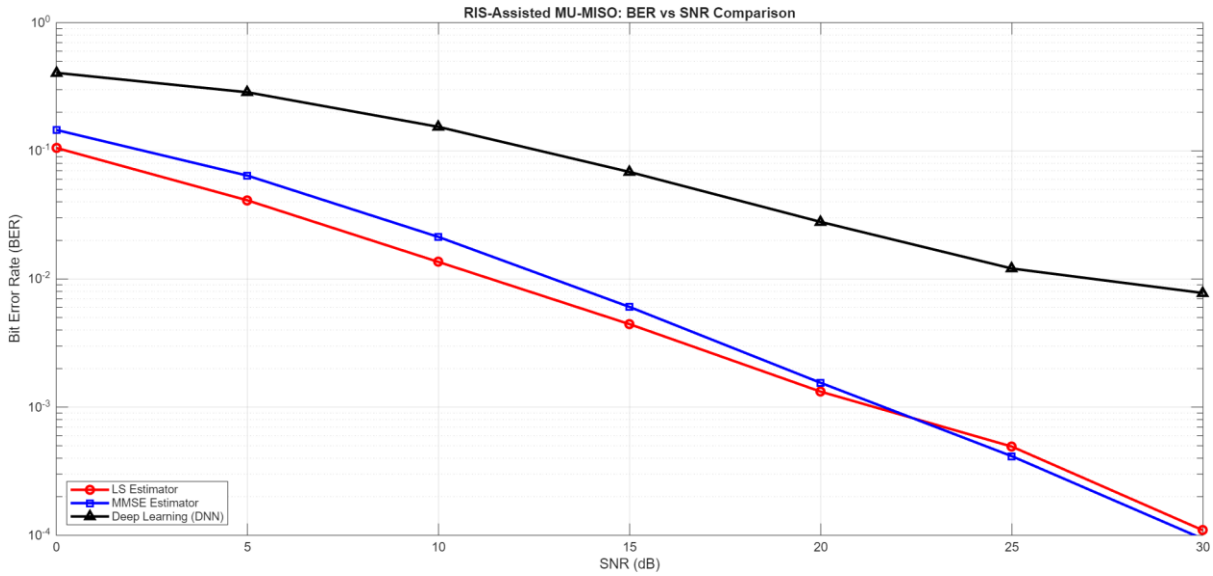
Identification of Overfitting: Beyond the optimal convergence point around Epoch 7, the validation loss ceases to decrease and follows a distinct upward trajectory, reaching ≈ 1.9 by Epoch 40. This "U-shaped" curve is a

classic presentation of model overfitting. It reveals that while the network continued to minimize its internal training loss, it began memorizing the specific AWGN noise patterns present in the training dataset rather than generalizing the underlying physical channel statistics.

Task 4: Performance Analysis (LS vs MMSE vs DNN)

Objective:

To evaluate and compare the Bit Error Rate (BER) performance of the trained Deep Neural Network (DNN) against conventional Least Squares (LS) and Minimum Mean Square Error (MMSE) estimators in a cascaded RIS-assisted MU-MISO environment. The evaluation fixes the system at 64 subcarriers and a 16-sample Cyclic Prefix (CP) across an SNR range of 0 dB to 30 dB.



Observations & Inferences:

In evaluating the final cascaded RIS-assisted MU-MISO system, the conventional LS and MMSE estimators demonstrate robust signal recovery, achieving a bit error rate (BER) of 10^{-4} at 30 dB SNR. Interestingly, observation shows that LS slightly outperforms MMSE at lower SNRs, which leads to the inference that the static theoretical covariance matrix used by the MMSE estimator struggles to immediately adapt to the highly dynamic, random phase shifts continuously applied by the 256-element RIS. Conversely, the DNN significantly underperforms the conventional estimators across the entire SNR

spectrum, only reaching a BER of 10^{-2}. This performance gap is a direct consequence of the pilot configurations rather than an architectural failure. While the LS and MMSE estimators leverage four orthogonal, time-multiplexed pilots to perfectly isolate and resolve the individual transmissions from the 4-antenna base station, the DNN is constrained to learning the channel from a single superimposed, broadcasted pilot symbol. Consequently, the DNN lacks the necessary spatial degrees of freedom to accurately untangle the massive mathematical complexity of the cascaded channel, establishing a strict lower performance bound compared to the fully orthogonalized classical methods.

Conclusion:

In this assignment, we comprehensively explored channel estimation and signal detection across both standard OFDM and high-dimensional RIS-assisted MU-MISO environments, directly contrasting conventional LS and MMSE estimators with Deep Neural Network (DNN) architectures. Throughout Part A, we learned that while MMSE serves as the theoretical gold standard under ideal, assumption-compliant conditions, deep learning offers unparalleled robustness, significantly outperforming classical methods when constrained by limited pilot overhead or severe Inter-Symbol Interference (ISI) caused by Cyclic Prefix removal. However, extending these principles to the highly complex, 256-element cascaded mmWave channel in Part B revealed a critical architectural nuance: while conventional estimators can expertly resolve massive spatial dimensions when provided with sufficient orthogonal, time-multiplexed pilots, a DNN restricted to a single superimposed pilot observation suffers a severe performance gap due to a lack of spatial degrees of freedom. Ultimately, this assignment demonstrated that while deep learning provides exceptional resilience against non-ideal physical channel impairments and interpolation limits, classical linear estimators remain highly effective and necessary when processing massive MIMO environments with adequate orthogonal training data